universe to be some positive number. (If you didn't, you'd be in favor of trying somehow to destroy the universe, which makes for a good supervillain origin story but not for good neighbors.)

It would follow that, if the universe has positive utility and our goal is to maximize utility, creating a new copy of the whole universe would be one of the most morally valorous actions you could possibly take. The right thing to do would then be to branch the wave function of the universe as often as possible. We could imagine building a quantum utility maximizing device (QUMaD), perhaps an apparatus that continually bounces electrons through a device that measures first their vertical spin, then their horizontal spin. Every time an electron undergoes either measurement, the universe branches in two, doubling the total utility of all universes. Having built QUMaD and turned it on, you would be the most moral person ever to live!

Something about this smells fishy, however. Turning on QUMaD has no impact whatsoever on the lives of people in this universe or any other. They don't even know the machine exists. Are we really sure it has such a morally praiseworthy effect?

Happily there are a couple of ways out of this puzzle. One is to deny the assumptions: maybe this kind of no-nonsense utilitarianism isn't the best moral theory. There is a long and honorable tradition of people inventing things that would nominally increase the utility of the universe, but don't resemble our moral intuitions whatsoever. (Robert Nozick imagined a "utility monster," a hypothetical being that was so good at experiencing pleasure that the most moral thing anyone could do would be to keep the monster as happy as possible, no matter who else might suffer thereby.) QUMaD is just another example along these lines. The simple idea of adding up utilities among different people doesn't always lead to the results we might initially have imagined.

But there's another solution, one that comports more directly with the Many-Worlds philosophy. When we talked about deriving the Born rule, we discussed how to apportion credences in conditions of self-locating uncertainty: you know the wave function of the universe, but you don't know which branch you are on. The answer was that your credences should be proportional to the weight of the branch—the corresponding amplitude, squared. This "weight" is a crucially important aspect of how we think about worlds in an Everettian picture. It's not just probability that goes that